Customer Shopping Behavior Analysis

A Data Analytics Project Report

This project analyzes customer shopping behavior using transactional data containing 3,900 purchase records across different product categories. The objective is to identify customer purchasing patterns, product preferences, spending behavior, subscription trends, and business opportunities through data analytics techniques.

1. Introduction

Understanding customer behavior is essential for modern businesses to improve marketing strategies, customer retention, and product planning. This project applies Python, PostgreSQL, and Power BI to transform raw shopping data into meaningful business insights.

2. Project Objectives

- Analyze customer spending patterns.
- Identify high-value customer segments.
- Understand product performance and customer preferences.
- Evaluate subscription and discount behavior.
- Develop an interactive business dashboard.

3. Dataset Description

The dataset contains:

- 3,900 rows
- 18 columns

Main features include:

- Customer demographics: Age, Gender, Location, Subscription Status
- Purchase information: Item Purchased, Category, Purchase Amount, Season, Size, Color
- Shopping behavior: Discounts, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

Missing data was found in the Review Rating column and handled during preprocessing.

4. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data loading and structure inspection:** Imported the dataset using pandas.

- **Statistical exploration using descriptive analysis:** Used df.info() to check structure and .describe() for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	Na
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	Na
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	Na
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	Na
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	Na
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	Na
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	Na

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

- **Missing value treatment using category-based median rating:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

- **Column standardization using snake_case naming:** Renamed columns to **snake case** for better readability and documentation.

- **Feature engineering by creating age_group and purchase_frequency_days:**

    - Created **age_group** column by binning customer ages.

    - Created **purchase_frequency_days** column from purchase data.

- **Removal of redundant promo_code_used information after consistency checking:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

- **Integration of cleaned data into PostgreSQL for SQL analysis:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

5. Data Analysis using SQL (Business Transactions)

PostgreSQL was used to answer important business questions:

Revenue by Gender:

The analysis compared revenue contribution between male and female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

High-Spending Discount Users:

Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	[illegible]

Total rows: 839 Query complete 00:00:(

Top Rated Products:

Products with the highest average review ratings were discovered.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Shipping Analysis:

Average purchase amounts were compared between shipping types.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Subscription Analysis:

Subscribers and non-subscribers were compared based on spending and revenue.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Customer Segmentation:

Customers were categorized into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Repeat Buyers Analysis:

The relationship between repeat purchasing behavior and subscription status was evaluated.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Age Group Revenue:

Revenue contribution was analyzed across different customer age groups.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Discount-Dependent Products:

Identified 5 products with the highest percentage of discounted purchases

	item_purchased text	discount_rate numeric
1	Socks	32.00
2	Blouse	33.00
3	Sandals	36.00
4	Skirt	38.00
5	Jacket	39.00

Product Category Analysis:

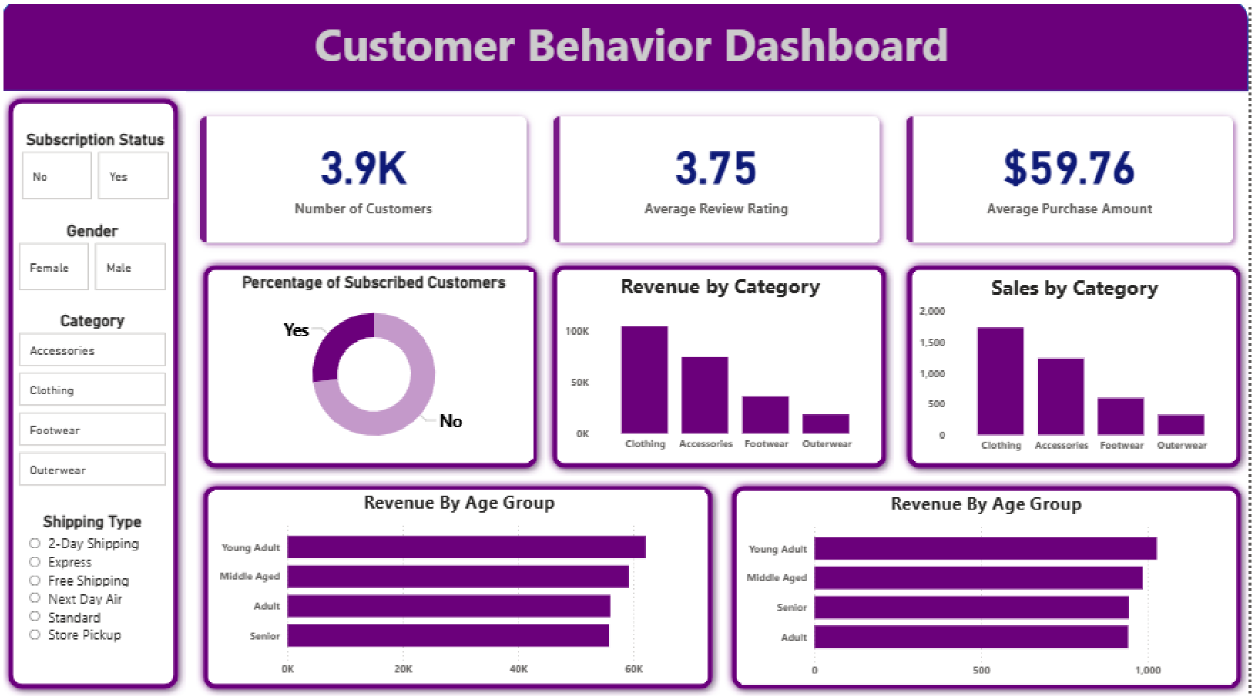
Top products within each category were identified.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

6. Power BI Dashboard

An interactive Power BI dashboard was created to visually present:

- Total customers
- Average review rating
- Average purchase amount
- Subscription percentage
- Revenue by category
- Sales by category
- Revenue by age group
- Customer filtering options



7. Key Findings

- Customer purchase behavior varies across demographic groups.
- Loyal customers represent the largest customer segment.
- Some products perform better based on ratings and sales volume.
- Discounts influence purchasing behavior for selected products.
- Subscription behavior provides opportunities for customer retention strategies.

8. Business Recommendations

Boost Subscriptions:

Offer exclusive benefits to increase subscription adoption.

Customer Loyalty Programs:

Reward repeats buyers to strengthen retention.

Review Discount Strategy:

Balance promotional discounts with profit margins.

Product Positioning:

Promote highly rated and best-selling products.

Targeted Marketing:

Focus campaigns on high-revenue customer groups.

9. Tools and Technologies

Python:

Data cleaning, exploration, and feature engineering.

PostgreSQL:

Business transaction analysis using SQL queries.

Power BI:

Interactive visualization and dashboard development.

10. Conclusion

This project demonstrates how data analytics can transform customer transaction data into actionable business insights. By combining Python, SQL, and Power BI, organizations can better understand customer behavior and make data-driven decisions.